

SCRB Sahayak (v2) — Comprehensive Project Context

> **Executive Briefing Document for AI Agents & Developers**

> **Repository:** `TESTKSPAI` / `KSPAI-main`

> **Domain:** Law Enforcement, Criminal Investigation Decision Support, Statutory Legal Compliance, Case Intelligence

> **Target User Base:** Karnataka State Police (KSP) officers (Constables, Investigating Officers, Inspectors, SP, DIG, IGP)

> **Primary Interfaces:** Bilingual React Web SPA (English & Kannada / ಕನ್ನಡ) + FastAPI Backend + PostgreSQL

1. Executive Summary & Mission

SCRB Sahayak v2 is a bilingual investigation-support and case-intelligence portal designed for modern police departments. Built as a prototype for the Karnataka State Police (KSP) Datathon using synthetic data, the platform solves key operational bottlenecks in police investigations:

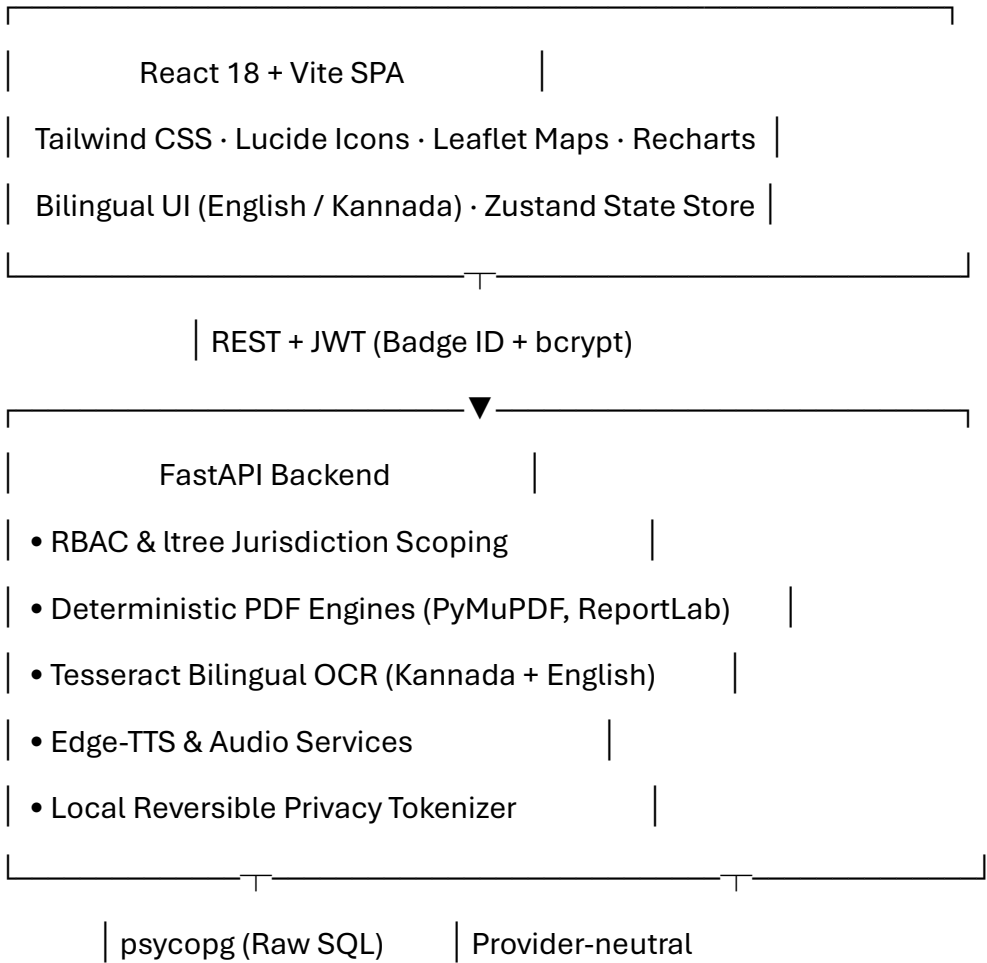
- Intelligent FIR & Case Intake:** Rapid intake of handwritten/scanned FIRs via bilingual OCR (English + Kannada) and entity extraction.
- Statutory Deadline Tracking (BNSS 187(3)):** Precision tracking of 60-day and 90-day custody/remand clocks per accused individual to prevent default bail.
- Deterministic Final-Report Builder (BNSS 193):** An officer-controlled, non-hallucinatory workflow for compiling formal charge sheets and closure reports (modeled after the Rajasthan IIF-IV standard).
- Cross-Case Linkage & Network Canvas:** Dynamic graph analytics discovering hidden links, shared phone numbers, vehicle registrations, and syndicates across police stations.
- Date-Wise Case Diary (Section 172 CrPC / Section 192 BNSS):** Page-numbered daily legal logs with chronological date-range PDF exports and assigned-IO write locks.

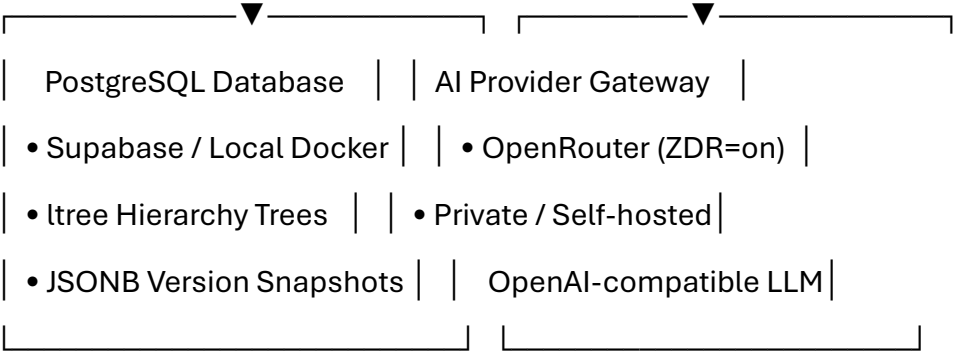
6. **Privacy-Controlled, Provider-Neutral AI Copilot:** Natural language case search and statistical breakdowns powered by an LLM with backend-enforced reversible pseudonymization and Zero Data Retention (ZDR).

> **Core Operating Principle:** **Deterministic for legal filings, AI-assisted for intelligence & search.** Critical legal documents and court filings are generated strictly by verified database records and code logic—never generated by generative LLM prompts.

2. High-Level System Architecture

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### Component Details

| Layer | Stack | Key Responsibilities |

|---|---|---

| **Frontend** | React 18, Vite, TypeScript, Tailwind CSS, Lucide React, Leaflet, Recharts, Framer Motion, Zustand | Responsive SPA, interactive canvas graph, audio record/playback, real-time filters, document viewers. |

| **Backend** | Python 3.11+, FastAPI, Pydantic v2, psycpg (raw async PostgreSQL), PyMuPDF, ReportLab | REST endpoints, authentication, deterministic validation state machines, PDF generation, OCR processing. |

| **Database** | PostgreSQL (Supabase or local Docker) with `ltree` extension | Relational data, hierarchical police command trees, immutable JSONB document revisions, audit logs. |

| **AI Gateway** | OpenRouter or private OpenAI-compatible endpoint (vLLM / Ollama) | Case briefing, natural language crime analytics, lead discovery, entity extraction. |

| **Privacy / Security** | In-memory tokenization, SHA-256 HMAC fingerprints | Reversible PII redaction, Zero Data Retention enforcement, metadata-only privacy audit logs. |

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## 3. Police Hierarchy & Jurisdiction Control (RBAC)

Access control strictly mirrors the real-world command structure of Indian police forces using PostgreSQL's `ltree` extension:

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State Command (Police IT / DGP)

- └─ Command Range (IGP - Multiple Districts)
 - └─ Officer District (DIG - Multi-district supervision)
 - └─ District Command (SP - Superintendent of Police)
 - └─ Sub-Division / Range (DySP)
 - └─ Police Station (Inspector / SHO)
 - └─ PSI (Investigating Officer - IO)
 - └─ Head Constable / Constable

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### ### Jurisdiction Rules:

- \* Every database query applies `jurisdiction\_filter\_sql` based on the authenticated officer's badge profile.
- \* Station officers can only view or query cases registered within their police station.
- \* SPs and senior supervisory officers can query cases across their subordinate districts or ranges.
- \* \*\*AI Copilot tools inherit the requesting officer's jurisdiction:\*\* An officer can never query or receive intelligence from cases outside their legal jurisdiction.

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## ## 4. Key Workflows & Modules

### ### A. Case Intake & Scanned FIR Ingestion

- \* **Bilingual OCR:** Ingests scanned PDF or image FIRs in Kannada or English using Tesseract.
- \* **Atomic FIR Serial Allocation ( `0018` ):** Allocates guaranteed non-colliding yearly FIR numbers ( `FIR/YYYY/XXXX` ) even under high concurrency.
- \* **Protected FIR Retention ( `0016` ):** Preserves original binary files with access audit tracking and download verification.
- \* **Distinct Person Records:** New person entities are created without naive name-matching; cross-case identity links remain officer-reviewed suggestions.

### ### B. Date-Wise Case Diary (Section 172 CrPC / Section 192 BNSS)

- \* Entries are strictly anchored to the Karnataka calendar date.
- \* **Per-Day Page Numbering:** Page numbering resets daily (Entry 1 on Aug 24 is Page 1, Entry 2 is Page 2; Entry 1 on Aug 25 starts back at Page 1).
- \* **Write Controls:** Strictly limited to the assigned Investigating Officer (IO).
- \* **Export:** Date-range filtered PDF export following official state police formatting.

### ### C. Statutory Custody Clocks (BNSS Section 187(3))

- \* **Legal Rule:** Default bail clocks run for **60 days** (lesser offences) or **90 days** (serious offences punishable with death, life imprisonment, or 10+ years).
- \* **Anchoring Point:** The custody clock is tied to a specific **accused person** and begins from the **first magistrate-authorized remand date/time** (NOT the FIR registration date).
- \* **Deadline Board:** Distinctly tracks remand clocks separately from general FIR-age progress.

### ### D. Criminal Network Canvas & Cross-Case Linkage

- \* Force-directed interactive canvas graph with zoom, pan, dragging, and cluster highlighting.
- \* Detects criminal syndicates through shared attributes:
  - \* Shared phone numbers, vehicle registrations, physical addresses.
  - \* Shared co-accused in prior cases across police stations.

\* \*\*Seed-First Neighbourhood Traversal ( ` 0017 ` ):\*\* Expands networks outward from an anchor case or person within authorized jurisdictions.

### ### E. Deterministic Final-Report Builder (BNSS Section 193)

\* Final reports (Charge Sheets or Closure Reports) are generated deterministically based on the Rajasthan IIF-IV standard:

1. \*\*Report Data Sources ( ` 0012 ` ):\*\* Verified parties, chronology, property, forensics, witness statements.

2. \*\*Draft Initialization:\*\* Pre-fills empty fields from verified case metadata without overwriting officer edits.

3. \*\*Validation Engine:\*\* Evaluates mandatory fields, legal section alignments, and contradictory entries.

4. \*\*Immutable Versioning:\*\* Creates distinct versions ( ` SOURCES\_REFRESHED ` , ` OFFICER\_EDITED ` , ` SUBMITTED ` , ` APPROVED ` ).

5. \*\*Supervisory Approval:\*\* Inspectors/SPs review, reject with feedback, or approve. Approved reports are locked against modifications.

6. \*\*ReportLab PDF Engine:\*\* Exports court-ready PDF documents.

### ### F. Provisional Investigation Playbooks ( ` 0013 ` )

\* Structured SOP checklists tailored by crime type (theft, robbery, homicide, cybercrime, narcotics).

\* Divided into standard investigative phases:

\* \*Immediate / Initial Actions (0–24 hours)\*

\* \*Early Actions / Evidence Collection (24–72 hours)\*

\* \*Forensic, Cyber & Interrogation Follow-up\*

\* \*Supervisory Review & Filing\*

\* Generates routine legal application drafts (e.g., CDR preservation requests, search warrant applications) deterministically from verified case parameters.

### ### G. Hotspot Analytics & Command Dashboard

\* Spatial crime mapping using Leaflet.

- \* Temporal breakdown by hour of day, weekday, month, and crime type.
- \* Early warning alerts trigger when anomalies (e.g., repeat crime clusters in a specific beat) are detected.

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## ## 5. AI Copilot & Privacy Gateway Architecture

The AI copilot provides natural language inquiry capabilities while maintaining strict data governance.

### ### A. Provider-Neutral Gateway ( `llm\_gateway.py` )

- \* Abstracts all LLM API calls behind a unified interface.
- \* Supports **OpenRouter** or any **OpenAI-compatible private endpoint** (local server, vLLM, Ollama).
- \* Enforces outbound character limits to prevent unexpected egress.

### ### B. Ephemeral Reversible Pseudonymization ( `ai\_privacy.py` )

- \* Before any prompt leaves the backend, personal identifiers are tokenized:

- \* `Anitha B.` → `[PERSON\_1]`
- \* `9845012345` → `[PHONE\_1]`
- \* `KA-01-AB-1234` → `[VEHICLE\_1]`
- \* `FIR/2026/0042` → `[FIR\_1]`

- \* The token mapping dictionary lives **only in ephemeral memory** for that single request turn.
- \* The model operates strictly on tokenized values; upon receiving completion or tool arguments, the backend replaces tokens back with original values before presentation to the officer.

### ### C. Zero Data Retention (ZDR) Enforcement

- \* By default, OpenRouter requests include ``provider: {"zdr": true}``.
- \* If a model endpoint does not support ZDR, the request fails closed rather than falling back to an unverified provider.

### ### D. Read-Only Function Calling Tools (``ai_tools.py``)

The LLM has access only to read-only database tools:

- \* ``search_cases``: Multi-parameter case search within officer's jurisdiction.
- \* ``get_crime_statistics``: SQL-computed aggregate counts, distributions, and trends.
- \* ``get_case_dossier``: Full case record with connected evidence and diary entries.
- \* ``run_case_intake``: Automated 24–72h investigative briefing for newly assigned cases.
- \* ``find_identity_matches``: Searches for potential suspect matches across past records.
- \* ``find_mo_similar_cases``: Discovers cases sharing Modus Operandi.
- \* ``suggest_legal_sections``: Maps crime narratives to BNS/IPC sections.
- \* ``get_deadline_risks``: Identifies impending 60/90-day remand expirations.

### ### E. System Prompt Guardrails (``ai.py``)

- \* **Strict Kannada Support:** If prompted in Kannada (or romanized Kannada), replies **must** be in authentic Kannada script (ಕನ್ನಡ ಲಿಪಿ).
- \* **Strict Grounding:** Facts must originate from tool outputs; guessing or hallucination is prohibited.
- \* **Impartiality:** MO patterns must only evaluate method, timing, location, and records—never religion, caste, or community.
- \* **Domain Lock:** Strictly limited to policing and investigation. Off-topic questions return exactly: ``"Ask relevant questions"``.

### ### F. Metadata-Only Privacy Audit (``0014``)

- \* ``AiRequestAudit`` logs officer ID, session ID, provider, model, token count, latency, and redaction category counts.



\* \*\*Never stores prompts, completions, narratives, or token maps\*\*, ensuring the audit log itself cannot become a vector for data leaks.

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## ## 6. Database Migration Sequence

The active schema evolves through sequential SQL migrations in  
`database/migrations/`:

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0003: Core relational schema (Case, Person, Station, District, Connection, AuditLog)

0004: Hierarchy & auth (ltree ranks, station-range paths, invitations, password resets)

0005: Command jurisdiction (CommandRange for IGPs, OfficerDistrict for DIGs)

0006: Early warnings & alerts (hotspots, read receipts, threshold scores)

0007: Case diary & evidence (daily diary entries, chain of custody, assigned IO)

0008: Case diary composite indexes

0009: BNSS 187(3) case custody clocks (accused-specific remand dates)

0010: Final report builder v1 (working report draft, versioning)

0011: Final report schema v2 (IIF-IV Rajasthan reference layout in JSONB)

0012: Case report sources (reusable party profiles, chronology, property, expert results)

0013: Provisional investigation playbooks (SOP task checklists & routine applications)

0014: AI privacy audit (metadata-only audit for LLM requests)

0015: Scalable case directory keyset indexes

0016: Protected original FIR document storage & audit tracking

0017: Seed-first network focus indexes

0018: Atomic FIR counter sequence

0019: Durable upload jobs & diary attachments

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## ## 7. Repository Layout Reference

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KSPAi-main/

└─ backend/

| └─ app/

| | └─ main.py # FastAPI application entry point & CORS

| | └─ config.py # Environment configurations (Settings)

| | └─ deps.py # JWT authentication dependency

| | └─ routers/

| | | └─ ai.py # Chatbot & copilot endpoints

| | | └─ cases.py # Case CRUD, dossier & party APIs

| | | └─ case_diary.py # Date-wise case diary & PDF export

| | | └─ final_reports.py # Section 193 Final Report workflow

| | | └─ network.py # Cross-case network graph endpoints

| | | └─ investigation_plans.py # SOP playbooks & document drafts

| | └─ services/

| | └─ ai_privacy.py # Ephemeral PII redaction/tokenization

| | └─ ai_tools.py # Read-only database tools for LLM

| | └─ llm_gateway.py # Provider-neutral AI client

| | └─ case_access.py # ltree jurisdiction filters & RBAC

| | └─ crime_queries.py # Deterministic SQL search & analytics

| | └─ final_reports.py # Final-report validation & state machine

└─ frontend/

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|   ├── src/
|   |   ├── pages/
|   |   |   ├── CaseDossierClient.tsx # Multi-tab detailed case view
|   |   |   ├── Network.tsx          # Interactive network graph canvas
|   |   |   ├── Deadlines.tsx        # BNSS remand clock & bail risk board
|   |   |   ├── DashboardClient.tsx  # Command metrics & crime trends
|   |   |   └── FirUpload.tsx        # Scanned FIR OCR & ingestion pipeline
|   |   └── components/scrb/
|   |       ├── ai-pill.tsx          # Floating AI assistant / quick-ask modal
|   |       ├── CaseDiaryTab.tsx     # Daily diary management
|   |       ├── ChargesheetEditor.tsx # Structured BNSS final report editor
|   |       └── InvestigationPlanTab.tsx # SOP checklist & routine drafts
|   ├── database/
|   |   ├── migrations/              # 0003 to 0019 SQL migrations
|   |   └── seed/seed.sql             # Synthetic demo dataset (30 cases, multi-district)
|   └── docs/                        # Architectural design documents & Phase reports
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8. Guidelines for AI Agents Modifying this Codebase

1. ****Do Not Generate Final Legal Reports via LLM:**** The BNSS 193 report builder is deterministic. Use `[` backend/app/services/final_reports.py` ](file:///c:/Users/ZMAGYEET/Downloads/KSPA i-main/KSPAi-main/backend/app/services/final_reports.py)` and ReportLab for PDF output.
2. ****Always Enforce Jurisdiction Filters:**** When querying cases, always pass the current ``officer`` object to

[` backend/app/services/case\_access.py`](file:///c:/Users/ZMAGYEET/Downloads/KSPAi-main/KSPAi-main/backend/app/services/case_access.py) to inject
`jurisdiction\_filter\_sql` .

3. ****Keep AI Tools Read-Only:**** The AI assistant provides leads and drafts. Identity matching confirmation, status changes, and case closures must be executed by authenticated human officers in the UI.

4. ****Preserve PII Redaction:**** When introducing any new external LLM call, ensure all user and case content is passed through `PrivacySanitizer` in
[` backend/app/services/ai\_privacy.py`](file:///c:/Users/ZMAGYEET/Downloads/KSPAi-main/KSPAi-main/backend/app/services/ai_privacy.py).

5. ****Accused-Specific Remand Clocks:**** Do not confuse FIR age with default bail timers. BNSS Section 187(3) custody clocks strictly begin from the date and time of the accused person's first magistrate-authorized remand.